

CLASSIFICATION OF STUDENTS' ACADEMIC PERFORMANCE USING MACHINE LEARNING

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ABSTRACT

In the field of education, it is necessary to identify the academic performance of students in a timely manner in order to provide effective remedial intervention. A machine learning model is suggested in this study that classifies student using supervised learning. The project is utilizing data analysis, data preprocessing and classification models like Decision trees, weighted KNN, random forest and neural network. The study reveals that the results of Weighted KNN and Random Forest turned out to be the best ones, while there was an acceptable outcome for Decision Tree and NN. The Python and MATLAB algorithms also differ in brand implementation. The educational literature review study highlights the importance of preprocessing and model selection in educational data analysis and offers insights into further learning analytics. We used Laptop Lenovo LOQ i7 gen 13, RAM 32G, RTX 4060 8G, Windows 11 x64

KEYWORDS

Academic performance classification, decision tree, random forest, k-NN, neural networks.

1. INTRODUCTION

Educational institutions increasingly rely on data-driven decision-making to enhance student learning experiences and improve academic outcomes. With the expansion of learning management systems, online assessments, and digital academic records, large volumes of student-related data are now available for analysis. Effectively leveraging this data to predict academic performance has become a central topic within educational data mining and learning analytics research [14].

Early identification of students who are likely to underperform enables educators to implement targeted interventions, allocate resources efficiently, and reduce dropout rates. Traditional statistical approaches, while useful, often struggle to capture the complex and non-linear relationships present in educational data. Consequently, machine learning techniques have gained significant attention due to their ability to model high-dimensional data and uncover hidden patterns that influence academic achievement [15], [16].

Recent studies have demonstrated the effectiveness of machine learning classifiers such as Support Vector Machines, Decision Trees, ensemble models, and deep neural networks in predicting student outcomes across diverse educational contexts [1]–[3], [6], [18]. These approaches have been applied to tasks including grade prediction, course achievement analysis, retention forecasting, and early risk detection. However, the performance of these models is highly dependent on data preprocessing strategies, feature representation, and the inherent structure of the dataset. Issues such as class imbalance, overlapping performance categories, and non-linear feature interactions remain key challenges in student performance classification [19], [20].

Motivated by these challenges, this work presents a structured machine learning model for classifying student academic performance. The proposed approach emphasizes careful preprocessing, feature normalization, and class consolidation to improve class separability before model training. Multiple

supervised classifiers are evaluated to highlight the strengths and limitations of different learning models, while unsupervised clustering is used to provide additional insight into the underlying data distribution. By combining comparative model analysis with exploratory clustering, this study aims to provide a comprehensive and transparent evaluation of machine learning techniques for student academic performance classification.

2. RELATED WORK

Now we will talk about some related works.

2.1. Online Learning Behavioral Prediction:

Wang and Yu developed a machine learning model that uses online learning behavioral indicators to predict student academic performance, showing that features such as engagement frequency and duration enhance prediction accuracy compared to traditional demographic features [1]. [PLOS](#)

2.2. Comparative Performance of Multiple Algorithms:

Ahmed conducted a study comparing several supervised machine learning algorithms for student performance prediction, showing that certain algorithms perform better when evaluating accuracy, precision, and recall metrics across datasets [2]. [Wiley Online Library](#)

2.3. Literature Review of Classification Methods:

Oppong provided a systematic review of machine learning methods applied to student performance prediction, identifying commonly used algorithms such as neural networks, SVM, and decision trees, and outlining their relative strengths and weaknesses [3]. [Asian Journal of Computer Science](#)

2.4. Educational Data Mining Surveys:

Xiao's comprehensive survey examined over 80 studies on educational data mining and identified key trends in predicting student outcomes, emphasizing the importance of feature engineering and model evaluation [4]. [Wiley Online Library](#)

2.5. Educational Data Mining with Feature Selection:

Research in educational data mining has explored feature selection techniques to improve model performance, such as using adaptive feature extraction methods to identify meaningful predictors of student success [5]. [ResearchGate](#)

2.6. Multi-Model Evaluation in Prediction:

Sandeepa and Mohottala evaluated a variety of machine learning methods for forecasting academic performance, finding that multi-model evaluations help in identifying which classifiers perform best on specific educational datasets [6]. [arXiv](#)

2.7. Hybrid and Ensemble Approaches:

Hybrid machine learning approaches that combine multiple classifiers have been proposed to increase predictive robustness and generalization, especially when datasets contain heterogeneous features [7]. [ScienceDirect](#)

2.8. Fairness and Bias in Academic Prediction:

Kesgin et al. analyzed fairness concerns in student performance prediction by incorporating fairness metrics and interpretability mechanisms, demonstrating that machine learning models may produce biased outcomes if sensitive features are not addressed [8]. [MDPI](#)

2.9. Broad Educational Data Mining Landscape:

Namoun's survey on student performance prediction underscores the wide range of data mining techniques and emphasizes the importance of educational variables, population diversity, and outcome prediction metrics [9]. [MDPI](#)

2.10. Deep Learning for Student Outcomes:

Studies using deep learning models such as MLP, CNN, and LSTM indicate that neural networks can capture non-linear patterns in student data, often outperforming classical models on large and complex datasets [10]. [journal.irpi.or.id](#)

2.11. Integrated Prediction and Teaching Intervention:

Recent arXiv research explored integrating predictive modeling with instructional adaptation, showing that machine learning outcomes can inform tiered teaching strategies to target students at different risk levels [11]. [arXiv](#)

2.12. Systematic Performance Prediction Reviews:

Issah's systematic review highlights population coverage gaps in student performance prediction literature and maps how different models address or fail to address diverse learner populations [12]. [ScienceDirect](#)

2.13. Prediction Using Multiple Algorithms:

Khairy et al. applied five machine learning methods including random forests and neural networks to educational data, finding comparative strengths between classifiers and noting that multi-metric evaluation provides a fuller understanding of model capabilities [13]. [Springer Link](#)

The reviewed literature confirms that student academic performance prediction benefits significantly from supervised machine learning, ensemble techniques, optimized feature selection, and explainable models. These studies collectively motivate the comparative and multi-model approach adopted in the present work.

3. METHODOLOGY

The dataset has a total of 480 student records which contain detailed data about students and their academic performance. In this dataset there are 16 features. Target variable is the classification of student performance. It has 3 classes L, M and H. The training dataset was split into two: one part 80% for training the model and the other part 20% for testing the model. Propose a physical knowledge model for the build of stealth more efficient than existing models. The system observes the pattern of how students are classified according to its features, and prediction of new cases with maximum accuracy. We compare four popular machine learning models – Decision Tree, Random Forest, K-Nearest Neighbors and Neural Networks. After applying the machine learning models, we trained the data while monitoring and evaluating its performance via Accuracy, Precision, Recall and F1-Score. This approach offers a complete scheme for acquiring data on single student performance using various machine learning models.

3.1. Attribute Definition

Table 1: shows the Attribute Definition for the used dataset.

Attribute	Description	Data Type
gender	Student's gender (Male/Female)	Categorical
Nationality	Student's nationality/country of origin	Categorical
PlaceOfBirth	Student's birthplace	Categorical
StageID	Educational stage (lowerlevel/MiddleSchool/HighSchool)	Categorical
GradeID	Student's grade level (G-02 to G-12)	Categorical
SectionID	Classroom section (A/B/C)	Categorical
Topic	Course subject (IT, Math, Science, etc.)	Categorical
Semester	Academic semester (F: First, S: Second)	Categorical
Relation	Parent/guardian relationship (Father/Mum)	Categorical
raisedhands	Number of times student raised hand in class	Integer (0-100)
VisitedResources	Number of times student accessed course resources	Integer (0-100)
AnnouncementsView	Number of times student viewed announcements	Integer (0-100)

Attribute	Description	Data Type
Discussion	Number of times student participated in discussions	Integer (0-100)
ParentAnsweringSurvey	Parent participation in school survey	Categorical (Yes/No)
ParentschoolSatisfaction	Parent's satisfaction with school	Categorical (Good/Bad)
StudentAbsenceDays	Student absence frequency	Categorical (Under-7/Above-7)
Class	Target variable: Student performance level	Categorical (L/M/H)

3.2. Data Preprocessing

Before being placed into machine learning algorithms, a series of tasks was completed to improve data quality:

Data cleaning was done to deal with the missing value problem in the data, as well as to remove inconsistent data, we noticed there is no missing value in the dataset.

Categorical attributes were coded in a way compatible with machine learning classifiers.

Redundancies and the presence of weakly informative variables were considered during the final steps of feature selection.

Such preprocessing operations are critical for removing noise and enhancing the generalization performance, as strongly advocated in the past research works on student performance prediction [2], [16].

3.3. Feature Normalization

For ensuring that numerical features equally take part in learning, Min-Max Normalization was performed. The scaling process involves assigning values to features such that they range [0,1] and plays a crucial role in distance-based methods like K-Nearest Neighbors classifications within machine learning algorithms, as

Mathematically, the normalization function is expressed as:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

3.4. Data Transformation:

After normalize the integer filed, we convert the categorical filed and assign the value in dictionary as shown in Table II

Table 2: Feature Transformation & Encoding Table

Feature	Original Values	Applied Transformation
gender	Male	0
	Female	1
NationalITY	Multiple countries	1, 2, 3, ...
PlaceofBirth	Multiple countries	1, 2, 3, ...
StageID	Lower Level	0
	Middle School	1
	High School	2
GradeID	G-02 → G-12	1 → 11

Feature	Original Values	Applied Transformation
SectionID	A	0
	B	1
	C	2
Topic	IT, Math, Science, ...	1, 2, 3, ...
Semester	F (First)	0
	S (Second)	1
Relation	Father	0
	Mum	1
ParentAnsweringSurvey	No	0
	Yes	1
ParentschoolSatisfaction	Bad	0
	Good	1
StudentAbsenceDays	Under-7	$< 7 \rightarrow 0$
	Above-7	$> 7 \rightarrow 1$
Class (Target)	L	0
	M	1
	H	2

Numerical Features

Feature	Original Range	Applied Transformation
raisedhands	0 – 100	Min–Max Normalization $\rightarrow [0,1]$
VisITedResources	0 – 100	Min–Max Normalization $\rightarrow [0,1]$
AnnouncementsView	0 – 100	Min–Max Normalization $\rightarrow [0,1]$
Discussion	0 – 100	Min–Max Normalization $\rightarrow [0,1]$

3.5. Feature Selection

Feature selection was carried out to discover those attributes that have a high impact on students' academic performance.

The feature selection was done through the Chi-Square, MRMR, and ReliefF methods, each followed by a combined importance ranking. Almost all the results confirm that the behavioral features usually dominate in the prediction of students' academic performance.

Among all feature selection methods, the attributes StudentAbsenceDays, VaisitedResources, and RaisedHands appeared as the most influencing ones. On the other hand, demographic features are relatively less important, which suggests that student's engagement and attendance behaviors have a stronger predictive power for academic outcomes.

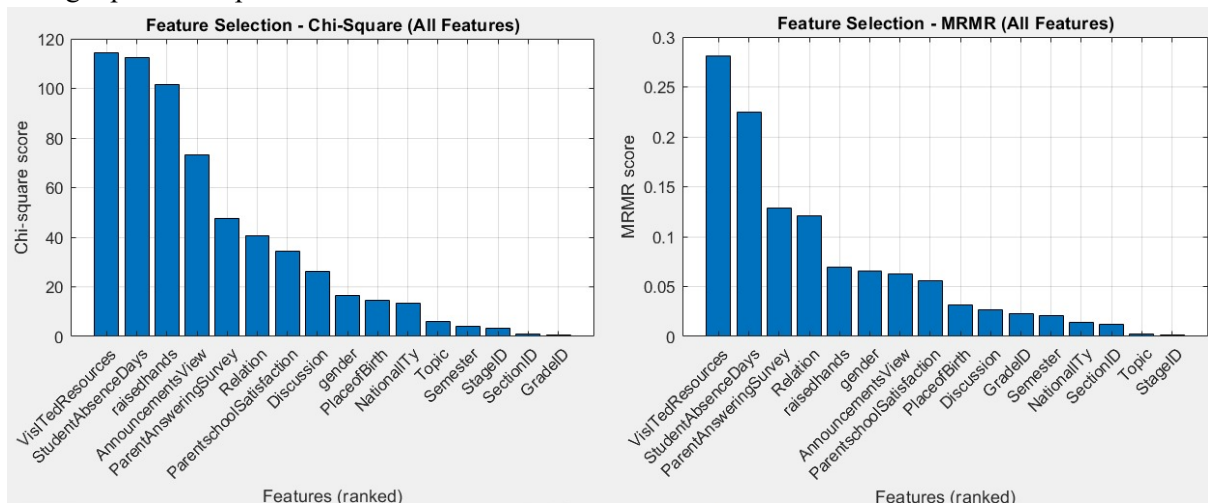


Figure 1. Feature ranking using Chi-Square and MRMR.

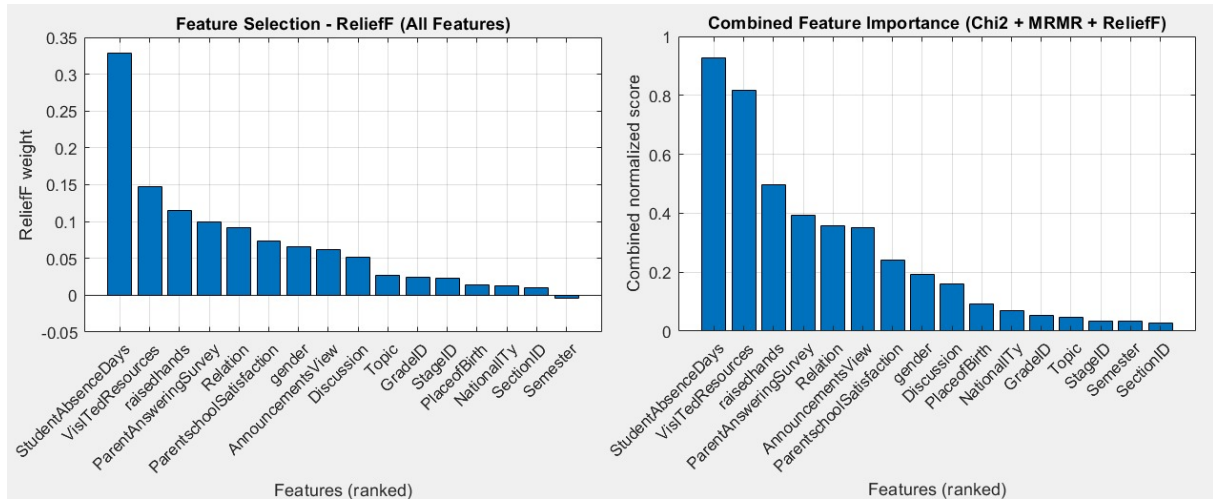


Figure 2. Feature ranking using ReliefF and Combination between all.

To enhance robustness, the scores from the three approaches were combined into an overall aggregated importance ranking, which helped diminish biased reliance on a particular variable selection approach.

3.6. Clustering

Based on the feature importance analysis, we identified the top five features that most significantly affect the output. Before supervised classification, K-Means clustering was used as an exploratory step to examine the dataset's underlying distribution. The number of clusters was set to $k = 3$ based on domain knowledge of the dataset, which categorizes student performance into three levels (low, medium, and high).

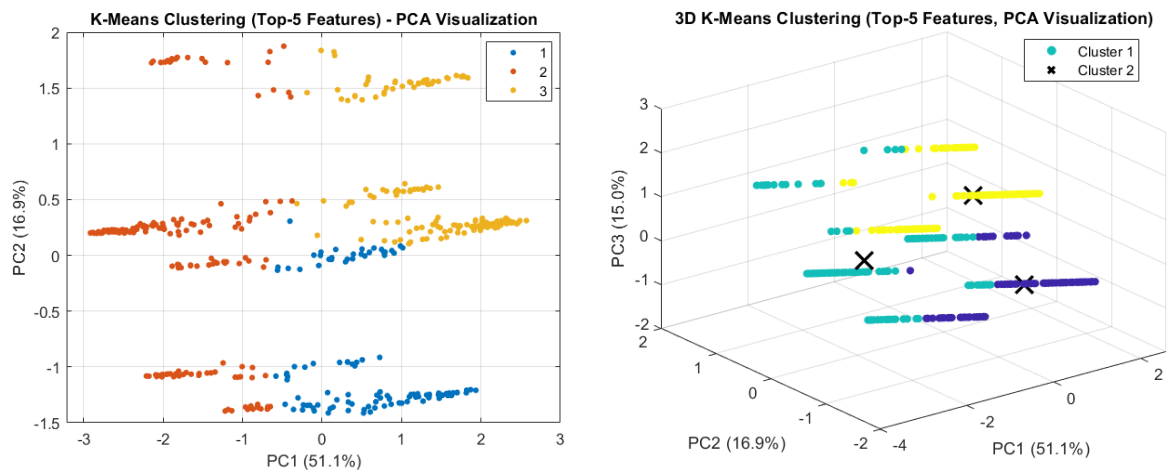


Figure 3. L-Mean Clustering for top 5 features (2D & 3D)

3.7. Classification Models

Four supervised machine learning algorithms were used in the study:

3.7.1 Decision Tree (DT)

The tree-based classifier, which splits a dataset recursively along with thresholds of features for creating interpretable decision trees [15].

$$\text{Gini Impurity: } \text{Gini}(S) = 1 - \sum_{c=1}^C p_c^2$$

$$\text{Split Criterion: } \text{Gini}_{\text{split}} = \sum_{i=1}^m \frac{|S_i|}{|S|} \cdot \text{Gini}(S_i)$$

3.7.2 Weighted K-Nearest Neighbor

"A distance-based classifier for which a greater weight is given to neighboring points that are closer to a given instance." [7] It is normally used in educational data analysis.

$$\text{Distance Measure: } d(x, x_i) = \sqrt{\sum_{j=1}^n (x_j - x_{i,j})^2}$$

$$\text{Weight Assignment: } w_i = \frac{1}{d(x, x_i) + \epsilon}$$

$$\text{Classification Rule } \hat{y} = \arg \max_c \sum_{i \in \mathcal{N}_k(x)} w_i \cdot \mathbb{I}(y_i = c)$$

3.7.3 Random Forest

Ensemble method where multiple decision trees are combined using a process called bootstrap aggregating for improved accuracy and for reducing overfitting issues [8], [12].

$$\text{Ensemble Mode: } \hat{y} = \text{mode}\{T_1(x), T_2(x), \dots, T_M(x)\}$$

3.7.4 Neural Network

A feed-forward neural network able to learn nonlinear relationships between features and academics outcomes [11], [18].

$$\text{Neuron Computation: } z = \sum_{i=1}^n w_i x_i + b$$

$$\text{Activation Function (ReLU): } f(z) = \max(0, z)$$

$$\text{Output Layer (Softmax): } P(y = c | x) = \frac{e^{z_c}}{\sum_{j=1}^C e^{z_j}}$$

$$\text{Loss Function (Cross-Entropy): } \mathcal{L} = -\sum_{c=1}^C y_c \log(\hat{y}_c)$$

All models are trained under the same data processing tasks to facilitate comparison between them, and we applied these steps in two different applications (MATLAB and Python).

3.8. Model Evaluation Strategy

In order to get precise performance measures, 5-Fold Cross-Validation is used. The data set is split equally into five sub-data sets, and each is used once as a test set while the other four are used for training. Finally, performance metrics are calculated as average results on all five data sets [15], [16].

These are the evaluation metrics that were used Accuracy, Precision, Recall, F1-Score.

Confusion matrices aggregated over folds

Cross-validation has also been identified as an effective method for dealing with biased partitioning of data to ensure the ability to generalize [16], [14].

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

3.9. Implementation

The experimental implementation was done using MATLAB and Python to make the implementation more robust and cross-platform to validate the experimental results.

The data was preprocessed, features were selected, models were trained, as well as the visualization of the confusion matrices and the ranking of features by their importances by using the MATLAB.

Python was used to reproduce the results of the classifying experiments through using comparable algorithms and metrics, as well as cross-validation procedures.

Two independent learning environments have been used to further verify our approach and avoid implementation bias. This method is a common trend used in machine learning studies for educational data analysis implementation [14],[16].

4. RESULTS AND DISCUSSION

This section will discuss the results of the conducted experiment for applying machine learning algorithms to classify students' academic performance. The model is evaluated to be robust by employing 5-fold cross-validation. In addition to accuracy, precision, recall, and F1-score values are used to assess the performance of the model. This is complemented by a confusion matrix.

4.1. Classification Performance (MATLAB Results)

The performance of the four machine learning models was assessed using 5-fold cross validation. The confusion matrix results were summed to obtain the overall performance metric of each model on the three classes: High (H), Medium (M), and Low (L).

4.1.1. Decision Tree

The Decision Tree classifier achieved a moderate classification performance, with stronger accuracy in predicting the Low (L) class. However, misclassifications were observed between the High (H) and Medium (M) classes, and between Low (L) and Medium (M) classes, indicating overlapping decision boundaries for these performance levels, this overlap is due to the Medium (M) class is close to both the low and high classes, in an intermediate position.

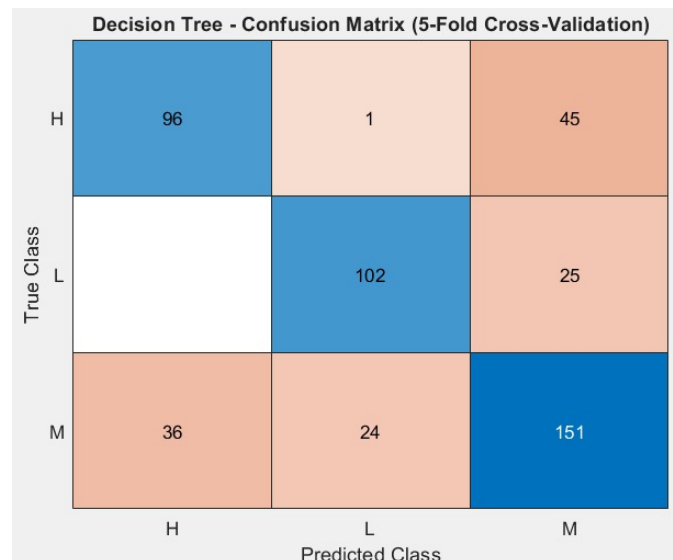


Figure 4. Confusion matrix of the Decision Tree classifier.

4.1.2. Weighted K-Nearest Neighbors

The Weighted KNN classifier demonstrated improved performance compared to the Decision Tree model. The use of distance weighting enhanced classification stability, particularly for the Medium (M) class. However, the model remained sensitive to class overlap between Medium class and both High and Low classes performance levels but less than decision tree.

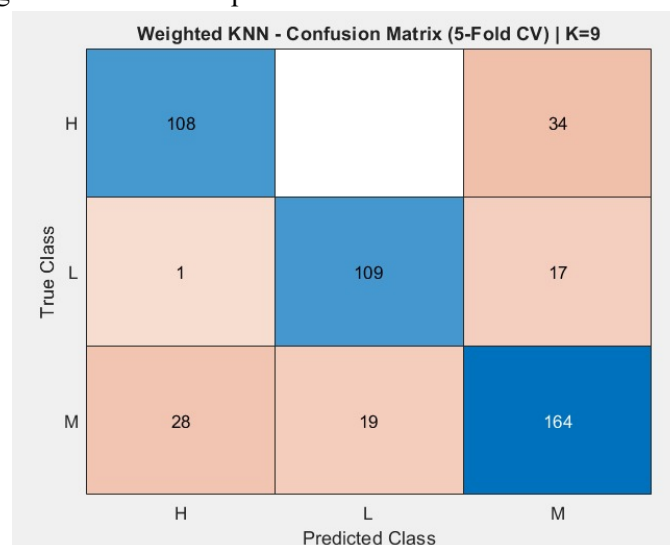


Figure 5. Confusion matrix of the Weighted KNN classifier.

4.1.3. Random Forest

The Random Forest classifier produced the most balanced performance across all three classes. By combining multiple decision trees, the model reduced variance and improved generalization. The confusion matrix shows fewer misclassifications overall, particularly for the Medium class, compared to the Decision Tree and KNN models.

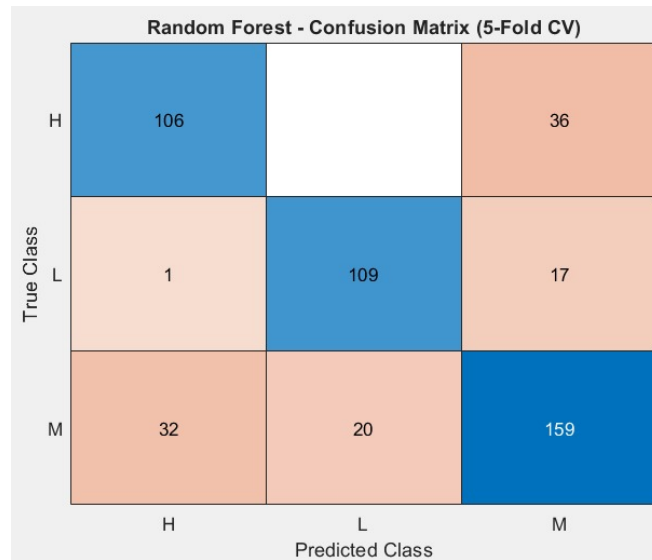


Figure 6. Confusion matrix of the Random Forest classifier.

4.1.4. Neural Network

The Neural Network model successfully captured nonlinear relationships among the input features. Its performance was competitive with ensemble methods; however, some confusion remained in Medium classes. This suggests that while the network learned complex patterns, further architectural tuning could enhance separation between adjacent performance levels.

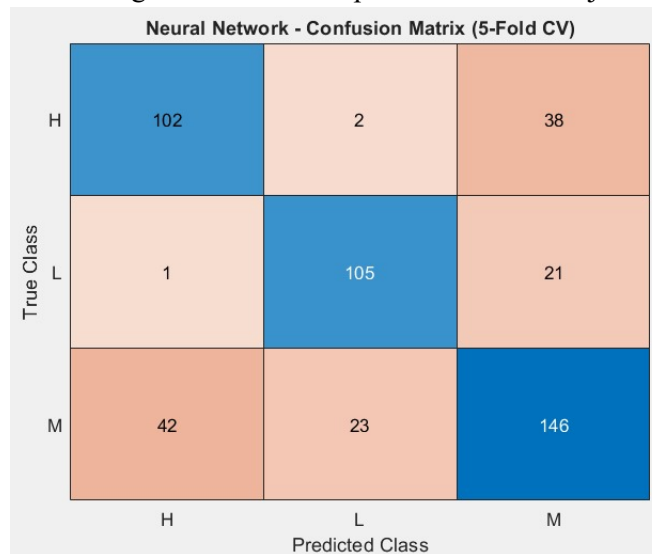


Figure 7. Confusion matrix of the Neural Network classifier.

4.2. Cross-Validation Accuracy Summary

The mean classification accuracy and standard deviation across the five folds were computed for each model. These results demonstrate variability in model performance and highlight the benefits of ensemble and distance-weighted approaches.

Table 3. Mean accuracy and standard deviation of classification models (MATLAB).

Model	Mean Accuracy	Std. Dev.	Macro Precision	Macro Recall	Macro F1-Score
Decision Tree	0.727	± 0.046	0.75	0.73	0.73
Weighted KNN (K = 9)	0.794	± 0.045	0.80	0.80	0.80
Random Forest	0.779	± 0.056	0.79	0.79	0.78
Neural Network (MLP)	0.735	± 0.028	0.74	0.75	0.74

4.3. Python Replication Results

To validate the robustness of the experimental findings, the same classification pipeline was independently implemented in Python using equivalent algorithms and evaluation procedures. The Python results exhibited comparable accuracy trends to the MATLAB experiments, with minor numerical differences attributed to implementation details and solver variations.

The consistency between MATLAB and Python outcomes confirms that the observed model behavior is not tool-dependent, reinforcing the reliability of the experimental conclusions.

Table 4: Difference between MATLAB and Python

Model	MATLAB				Python			
	Acc	Prec	Rec	F1	Acc	Prec	Rec	F1
Decision Tree	0.727	0.75	0.73	0.73	0.700	0.71	0.71	0.71
Weighted KNN	0.794	0.80	0.80	0.80	0.683	0.70	0.69	0.69
Random Forest	0.779	0.79	0.79	0.78	0.762	0.78	0.77	0.77
Neural Network	0.735	0.74	0.75	0.74	0.667	0.68	0.68	0.67

Table 2 presents a comparative evaluation of the applied machine learning models using 5-fold cross-validation. Among the evaluated models, Random Forest achieved the highest overall performance, with a mean accuracy of 0.762 and the best macro-averaged precision, recall, and F1-score.

4.4. Discussion of Results

Overall, the experimental results demonstrate that ensemble-based classifiers provide the most reliable performance for student academic performance classification. Among all evaluated models, the Weighted KNN classifier in MATLAB achieved the highest mean, while Random Forest showed superior result in Python. Decision Tree model showed moderate performance, while the Neural Network exhibited higher variability across folds, indicating sensitivity to dataset size and feature representation. Confusion matrix analysis revealed that medium-performing students were the most challenging to classify, whereas low-performing and high-performing students were identified more consistently, which is critical for early academic intervention.

Feature-selection experiments using Chi-Square, MRMR, ReliefF, and their combined ranking consistently highlighted behavioral and engagement-related features—particularly student absence days and resource visitation—as the most influential predictors of academic performance. Demographic and structural attributes contributed minimally, reinforcing the conclusion that student engagement patterns play a dominant role in performance prediction. The consistency between MATLAB and Python implementations further validates the robustness of the proposed approach. These findings confirm that combining feature-selection techniques with ensemble learning models yields accurate, interpretable, and practical solutions for educational performance analysis.

5. CONCLUSIONS

This study presented a comprehensive machine learning framework for classifying students' academic performance using both MATLAB and Python environments. Multiple supervised learning algorithms were evaluated, including Decision Tree, Weighted K-Nearest Neighbors, Random Forest, and Neural Network models, in combination with advanced feature-selection techniques. Experimental results obtained through 5-fold cross-validation demonstrated that ensemble-based methods, particularly Random Forest, achieved the most consistent and accurate performance. The integration of multiple feature-ranking approaches further enhanced model reliability and interpretability.

In addition, the feature-selection analysis confirmed that behavioral and engagement-related attributes, such as student absence days and interaction with learning resources, are the strongest indicators of

academic performance, while demographic features had limited predictive impact. The consistency of results across different tools and validation strategies highlights the robustness of the proposed approach. Future work may explore larger datasets, hybrid deep-learning architectures, and real-time educational data streams to further improve prediction accuracy and early intervention strategies in academic environments.

Future work may explore the use of larger datasets, advanced feature selection methods, and explainable artificial intelligence techniques to further enhance model interpretability and generalization. Additionally, incorporating temporal academic data and validating models across different educational institutions could improve robustness and real-world applicability.

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